

COURSE TITLE : THERAPEUTIC EQUIPMENTS
COURSE CODE : 5241
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Pacemakers, Defibrillators and Stimulators	13
2	Diathermy, Laser and Lithotripsy	13
3	ICU and operation theatre equipments	13
4	Drug delivery, Endoscopy and Radiotherapy	13
TOTAL		52

Course General Outcome:

MODULE	G.O.	ON COMPLETION OF THE STUDY OF THIS MODULE THE STUDENT WILL BE ABLE:
1	1	To understand the principle and working of cardiac pacemakers
	2	To Understand the principle of defibrillators and electrical stimulators
2	3	To comprehend the working of different methods of diathermy
	4	To Understand the principle of biomedical lasers and their applications
	5	To understand lithotripsy
3	6	To understand ICU equipments
	7	To Comprehend the working of operation theatre equipments
4	8	To comprehend the working of drug infusion systems and endoscopes
	9	To understand radiotherapy

G.O.: General Outcome

On completion of the study the student will be able

MODULE I PACEMAKERS, DEFIBRILLATORS AND STIMULATORS

- 1.1.0 To understand the principle and working of cardiac pacemakers
 - 1.1.1 To explain the working of cardiac pacemakers.
 - 1.1.2 To state the design requirements of cardiac pacemakers
 - 1.1.3 To explain the power sources of cardiac pacemakers.
 - 1.1.4 To explain the types of external and implantable pacing modes.
 - 1.1.5 To describe the working of ventricular synchronous demand pacemakers
 - 1.1.6 To describe programmable pacemakers.
- 1.2.0 To Understand the principle of defibrillators and electrical stimulators
 - 1.2.1 To explain the principle of defibrillators.
 - 1.2.2 To explain the working and output waveforms of DC defibrillators.
 - 1.2.3 To describe defibrillator with synchronizer, cardioverter

- 1.2.4 To describe Automatic Implantable Defibrillators.
- 1.2.5 To explain the types of pain relief through electrical stimulation.

MODULE II DIATHERMY, LASER AND LITHOTRIPSY

- 2.1.0 To comprehend the working of different methods of diathermy
 - 2.1.1 To explain the principle ,working and modes of operation of surgical diathermy.
 - 2.1.2 To describe short wave diathermy
 - 2.1.3 To describe microwave diathermy
 - 2.1.4 To describe ultrasonic diathermy
- 2.2.0 To Understand the principle of biomedical lasers and their applications
 - 2.2.1 To explain the principle, basic operation and applications of biomedical lasers.
 - 2.2.2 To list the types of biomedical lasers.
- 2.3.0 To understand lithotripsy
 - 2.3.1 To explain the principle of lithotripsy
 - 2.3.2 To explain the principle and operation of extra corporeal shock wave lithotripter.
 - 2.3.3 To explain the principle and operation of ultrasonic lithotripter.

MODULE III ICU AND OPERATION THEATRE EQUIPMENTS

- 3.1.0 To understand ICU equipments
 - 3.1.1 To explain the working of bedside monitors.
 - 3.1.2 To explain the working of centralized patient monitors.
 - 3.1.3 To describe the principle, types and modes of operation of ventilators.
 - 3.1.4 To explain the inspiratory phase, cycling mechanisms and expiratory phase of ventilators.
 - 3.1.5 To describe a typical ventilator.
- 3.2.0 To Comprehend the working of operation theatre equipments
 - 3.2.1 To describe the block diagram of anaesthesia machine.
 - 3.2.2 To explain the principle and need of cardiopulmonary bypass.
 - 3.2.3 To describe the operation of types of oxygenators and pumps used in heart lung machine.
 - 3.2.4 To explain the operation of the block diagram of heart lung machine.
 - 3.2.5 To illustrate the effect of cardiopulmonary bypass on vital organs of human body.

MODULE IV DRUG DELIVERY, ENDOSCOPY AND RADIOTHERAPY

- 4.1.0 To comprehend the working of drug infusion systems and endoscopes
 - 4.1.1 To explain the principle of various types of drug infusion systems
 - 4.1.2 To explain the working and applications of syringe infusion pump.
 - 4.1.3 To explain the working and applications of peristaltic and rotary infusion pump.
 - 4.1.4 To explain the principle and operation of endoscopy.
 - 4.1.5 To describe the parts of an endoscope .
 - 4.1.6 To explain different types of endoscopy with applications.
- 4.2.0 To understand radiotherapy.
 - 4.2.1 To describe the isodose chart and biological effects of radio therapy.

- 4.2.2 To explain the principle and operation of Co-60 radiotherapy unit
- 4.2.3 To explain the principle and operation of LINACC.
- 4.2.4 To list the various applications of radiotherapy.

CONTENTS

MODULE I _PRINCIPLE AND WORKING OF CARDIAC PACEMAKERS

Working of cardiac pacemakers- design requirements of cardiac pacemakers- Power sources of cardiac pacemakers-Pacing modes-Working of ventricular synchronous demand pacemakers-Programmable pacemakers.

Principle of defibrillators and electrical stimulators

Principle of defibrillators- working and output waveforms of DC defibrillators- Defibrillator with synchronizer, cardioverter -Automatic Implantable Defibrillators- Types of pain relief through electrical stimulation.

MODULE II _WORKING OF DIFFERENT METHODS OF DIATHERMY

Principle ,working and modes of operation of surgical diathermy- Short wave diathermy- Microwave diathermy- Ultrasonic diathermy

Principle of biomedical lasers and their applications

Principle, operation and applications of biomedical lasers-types

Lithotripsy

Principle of lithotripsy- Principle and operation of extra corporeal shock wave lithotripter and ultrasonic lithotripter

MODULE III ICU EQUIPMENTS

Working of bedside monitors, Centralized patient monitors- Principle, types and modes of operation of ventilators- Inspiratory phase- cycling mechanisms - Expiratory phase of ventilators- A typical ventilator.

Working of operation theatre equipments

Block diagram of anaesthesia machine- Principle and need of cardiopulmonary bypass - Operation of types of oxygenators and pumps used in heart lung machine-Operation of the block diagram of heart lung machine- Effect of cardiopulmonary bypass on vital organs of human body.

MODULE IV WORKING OF DRUG INFUSION SYSTEMS AND ENDOSCOPES

Principle of various types of drug infusion systems - Working and applications of syringe infusion pump, peristaltic and rotary infusion pump- Principle and operation of endoscopy- Parts of an endoscope - Types of endoscopy with applications.

Radiotherapy.

Isodose chart and biological effects of radio therapy- Principle and operation of Co-60 radiotherapy unit, LINACC- Applications of radiotherapy.

TEXT BOOK

1. Handbook of Biomedical Instrumentation- RS Khandpur- Tata McGraw-Hill Pub., 2006
2. Biomedical Instrumentation –Arumugam- Anuradha Publications, 1994

REFERENCES

- 1.Introduction to biomedical equipment technology- Carr & Brown- Pearson Education India
2. Encyclopaedia of medical devices and instruments-John G Webster- Wiley, 1988
- 3.Biomedical instruments and measurements-Leslie cromwell- Prentice-Hall, 2011